

Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. Cappuccino is an Italian style coffee topped with a layer of frothy milk (Figure 1.1).



Figure 1.1

Frothy milk is made by bubbling steam through milk, which is held in a metallic jug (Figure 1.2). Steam is ejected from the steam wand of a cappuccino machine (Figure 1.3).

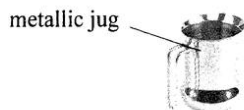


Figure 1.2



Figure 1.3

Given: specific latent heat of vaporization of water = $2.26 \times 10^6 \text{ J kg}^{-1}$
 specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$
 specific heat capacity of steam = $2000 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$
 specific heat capacity of milk = $3900 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

- (a) Calculate the total amount of heat released when 20 g of steam at 110°C cools to 100°C and condenses to water at 100°C . (3 marks)

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- (b) 20 g of steam at 110°C is bubbled through 200 g of milk at 15°C to make frothy milk. Using the result in (a), estimate the temperature of the frothy milk. (2 marks)

Answers written in the margins will not be marked.

- (c) Would the actual temperature of frothy milk be higher than, equal to or lower than the value found in (b)? Explain. (2 marks)

Answers written in the margins will not be marked.

Book 1 section 2 Heat Capacity: DSE long questions

2012 Q1 (page 3)

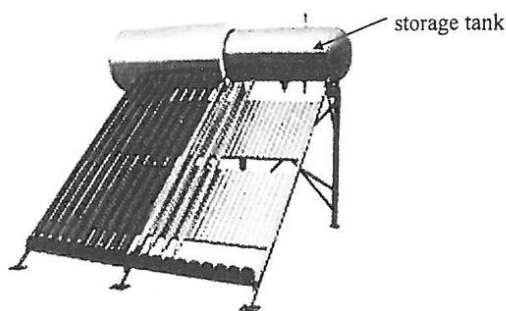
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Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

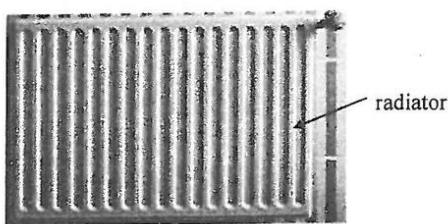
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Figure 1.1



A solar water heater shown in Figure 1.1 is installed on the rooftop of a house. During the day, the heater heats up 1.5 m^3 of water to 80°C . At night, the hot water in the storage tank is circulated to the radiators (see Figure 1.2) in different rooms of the house to keep the rooms warm.

Figure 1.2



Given: density of water = 1000 kg m^{-3}
specific heat capacity of water = $4200 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$

- (a) Given that 15% of the energy is lost during the transfer of water, how much heat can be released from the system to the rooms when the water temperature drops to 60°C ? (3 marks)

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- (b) Given that during night time the radiators maintain an average output power of 4.5 kW, how long can the radiators maintain this average power until the water temperature in the system drops to 60 °C ? Give your answer in hours. (2 marks)

Answers written in the margins will not be marked.

- (c) The rate of heat released by the solar water heating system during the time period calculated in (b) is in fact not constant and gradually drops. Explain why this is so. (1 mark)

Answers written in the margins will not be marked.

Book 1 section 2 Heat Capacity: DSE long questions

2013 Q1 (page 3)

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Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

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Figure 1.1

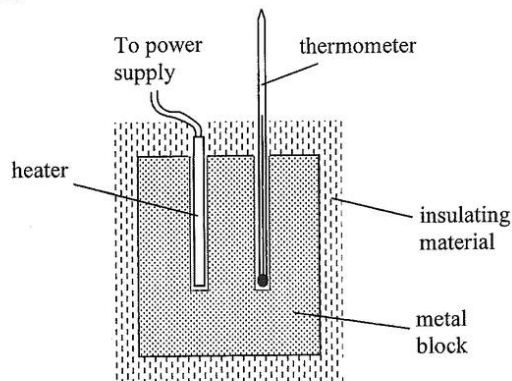
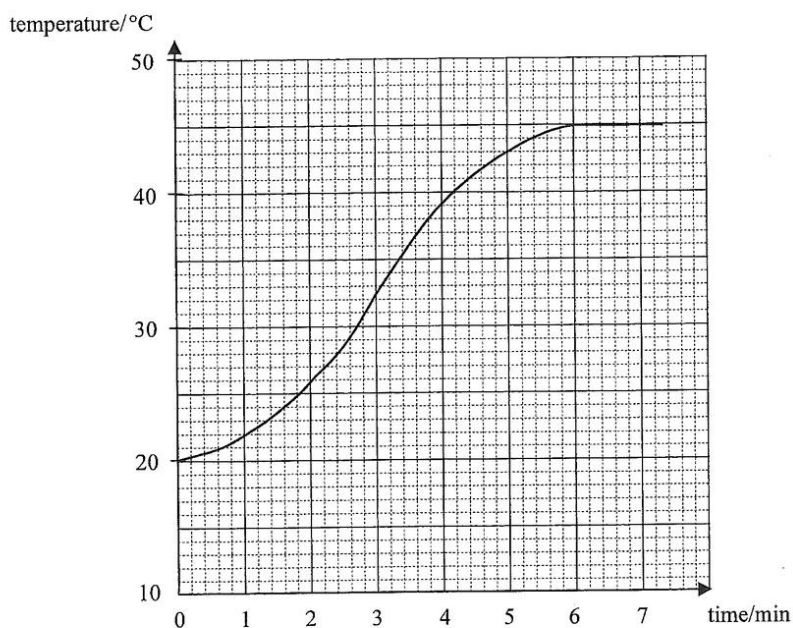


Figure 1.1 shows an experimental set-up to find the specific heat capacity of a metal. The metal block is wrapped by insulating material. A heater is connected to a power supply. It is switched on when the temperature of the metal block is 20°C and then switched off when the temperature reaches 43°C . The graph below shows the variation of the temperature of the metal block with time.



(a) Use the graph to find the duration that the heater is switched on. (1 mark)

(b) After the heater is switched off, the temperature of the metal block continues to rise for a while. Explain why. (1 mark)

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- (c) Given: mass of the metal block = 0.80 kg
heater voltage = 12 V
heater current = 4.0 A

- (i) By considering the maximum temperature rise of the metal block, calculate the specific heat capacity of the metal as found from the experiment. (2 marks)

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- (ii) Would your result be the same, higher, or lower than the actual value of the specific heat capacity of the metal? Explain. (2 marks)

Answers written in the margins will not be marked.

- (d) This method is not suitable for measuring the specific heat capacity of a glass block. Explain. (1 mark)

Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. The following experimental items are provided for estimating the specific heat capacity of bronze c_b :

- a bronze sphere of mass 0.80 kg hung with a thread at room temperature T_0
- a polystyrene cup containing 0.50 kg of water at room temperature T_0
- a water bath maintained at 80 °C
- a thermometer
- a stirrer
- a towel

(a) Describe the procedures of the experiment and state **TWO** experimental precautions to be taken. Write down an equation for finding c_b .

Given: specific heat capacity of water = 4200 J kg⁻¹ °C⁻¹

(6 marks)

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(b) The value of c_b found in the experiment in (a) is lower than the actual value. Explain. (2 marks)

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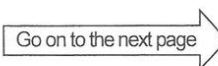
Book 1 section 2 Heat Capacity: DSE long questions

2016 Q1 (page 3)

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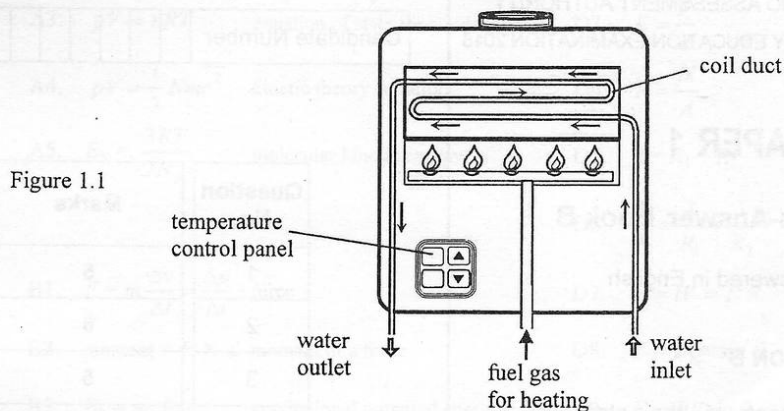
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Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. Figure 1.1 shows a domestic water heater. Tap water entering the heater is heated when it passes through the coil duct and exits as hot water at a certain temperature.



On a certain day in winter, the temperature of tap water is 15°C . When the heater is in use, it delivers 6 kg of hot water at a temperature of 50°C in 1 minute . Assume that there is no heat exchange between the heater and the surroundings. Given: specific heat capacity of water = $4200\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$

- (a) Estimate the power supplied to the tap water by the heater. (3 marks)

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- (b) Assuming the power estimated in (a) remains unchanged, determine the flow rate of tap water into the heater, in kg per minute , such that hot water at a temperature of 40°C is delivered by the heater. (2 marks)

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Book 1 section 2 Heat Capacity: DSE long questions

2018 Q1 (page 2)

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Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. (a) An insulated container of negligible heat capacity contains 1.5 kg of tea at a temperature of 60 °C.
- (i) What mass of ice at 0 °C should be added to the tea so that the final temperature of the mixture is lowered to 10 °C ? Assume that the specific heat capacity of tea is the same as that of water. (3 marks)
- Given: specific latent heat of fusion of ice = $3.34 \times 10^5 \text{ J kg}^{-1}$
 specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

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- (ii) If the heat capacity of the container is **not** negligible, explain whether more ice, less ice or the same amount of ice is needed to obtain the final temperature of 10 °C. (2 marks)

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- (b) Some ice cream at $-10\text{ }^{\circ}\text{C}$ is put into a 'thermal bag', of which the inner layer is made of polyethylene foam coated with aluminium foil. The bag is also equipped with a zipper at the top.



The thermal bag is then brought outdoors on a hot sunny day.

- (i) Referring to the heat transfer processes, explain ONE feature of this bag that helps keep the ice cream at a low temperature. (1 mark)

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- (ii) Suggest ONE modification to this bag that would enhance its ability to keep things stored inside at a low temperature. (1 mark)

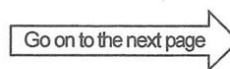
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Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. In a restaurant, 'wontons in soup' is prepared by putting 5 pieces of cooked wonton at 4°C into a bowl with 0.60 kg of soup at temperature 96°C .

Given: average mass of each piece of wonton = 0.02 kg
 specific heat capacity of wonton = $3300\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$
 specific heat capacity of soup = $4200\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$

- (a) Find the final temperature of the mixture. Assume that the heat capacity of the bowl and the heat loss to the surroundings are negligible. (2 marks)

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- (b) The soup in (a) is taken from a metallic container of heat capacity $2000\text{ J }^{\circ}\text{C}^{-1}$ containing 16 kg of soup maintained at 96°C by an immersion heater.

- (i) Why does that energy have to be supplied by the heater to keep the soup at 96°C ? (1 mark)

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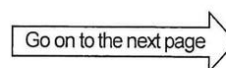
- (ii) A student used the following method to find the heater's operating power P : remove the heater from the container and record the temperature of the 16 kg of soup after 10 minutes. It is found that the temperature has dropped 9°C . Estimate P . (3 marks)

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- (iii) If the student repeats the measurement after another 10 minutes, would the corresponding temperature drop be larger than, equal to or smaller than 9°C ? Explain. (2 marks)

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1. A 150 W immersion heater is used to keep the water in a large beaker boiling under standard atmospheric pressure. In 5 minutes, 16 g of water boils away. Neglect any heat loss to surroundings.

(a) Find the specific latent heat of vaporization of water, l . (2 marks)

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A student puts a small metal sphere in the boiling water. After a few minutes, the sphere is quickly transferred to a polystyrene cup containing 100 g of water at a temperature of 20 °C. The cup of water is stirred gently and its highest temperature attained is 22 °C.

Given: specific heat capacity of water = 4200 J kg⁻¹ °C⁻¹

(b) Estimate the heat capacity C of the metal sphere. (2 marks)

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(c) In fact the sphere has carried with it some boiling water to the cup of water. Referring to this fact, explain whether the true value of C is higher or lower than the value calculated in (b). (2 marks)

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- (d) In order to reduce the error contributed by the polystyrene cup, another student suggests repeating the measurements using a copper cup of similar shape and size. Explain whether the suggestion is justified. (2 marks)

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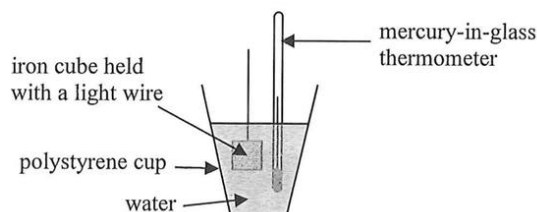
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Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. An iron cube of mass 200 g is heated to a few hundred degree Celsius (say at $T^{\circ}\text{C}$). It is then quickly transferred into a polystyrene cup containing 600 g of water. The cube is totally immersed in the water as shown in Figure 1.1. The temperature of the heated iron cube can be estimated in this experiment.

Figure 1.1



Assume that no heat is lost to the surroundings and neglect the heat capacity of the polystyrene cup.

- (a) The temperature of water recorded by the thermometer rises from 25°C to 33°C . Estimate T .
Given: The specific heat capacities of iron and water are $450 \text{ J kg}^{-1} ^{\circ}\text{C}^{-1}$ and $4200 \text{ J kg}^{-1} ^{\circ}\text{C}^{-1}$ respectively. (3 marks)

Answers written in the margins will not be marked.

- (b) What would happen to the water when this iron cube of high temperature just touches the water surface? (1 mark)

Answers written in the margins will not be marked.

- (c) Explain why the actual temperature of the iron cube should be higher than that estimated in (a). (2 marks)

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- (d) (i) A mercury-in-glass thermometer is not suitable for measuring the temperature of the heated iron cube **directly**. Why? (1 mark)

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- (ii) Suggest a kind of thermometer that can directly measure the temperature of this heated cube. (1 mark)

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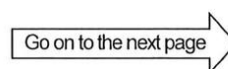
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Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided. Take $g = 9.81 \text{ m s}^{-2}$.

1. An empty glass cup is placed on a table under room temperature of 23°C . 400 g of water at 100°C is poured into the cup. After a moment, the temperature of water in the cup is found to be 93°C .

(a) State a cause for the drop of water temperature other than heat loss to the surrounding air. (1 mark)

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(b) Estimate the heat capacity H , in $\text{J }^\circ\text{C}^{-1}$, of the glass cup. (2 marks)
Given: specific heat capacity of water = $4200 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$

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(c) (i) What is the kind of electromagnetic waves, other than visible light, mainly emitted from this cup of hot water? (1 mark)

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(ii) Name a device in daily life which makes use of this kind of electromagnetic waves. (1 mark)

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Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided. Take $g = 9.81 \text{ m s}^{-2}$.

1. A mountaineer arrives at a location X , which is 3000 m above sea level. At X , the melting point of ice is 0°C and the boiling point of water is 90°C . The mountaineer needs to heat 0.50 kg of melting ice to its boiling point using a heater.

- (a) Estimate the energy required. (2 marks)

Given: specific latent heat of fusion of ice = $3.34 \times 10^5 \text{ J kg}^{-1}$
specific heat capacity of water = $4200 \text{ J kg}^{-1}^\circ\text{C}^{-1}$

Answers written in the margins will not be marked.

- (b) The power output of the heater is 2 kW. Estimate the time required. (2 marks)

Answers written in the margins will not be marked.

- (c) Explain why the temperature of water remains at the boiling point after the water starts to boil, even when the heater is still on. (1 mark)

Answers written in the margins will not be marked.

Book 1 section 2 Heat Capacity: DSE long questions

2026 Q1 (page 2)

Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided. Take $g = 9.81 \text{ m s}^{-2}$.

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*2. A sealed syringe contains neon gas of volume 5.0 cm^3 and pressure $1.0 \times 10^5 \text{ Pa}$. The gas is then compressed slowly to 2.0 cm^3 while the temperature remains at 27°C . Take the neon gas as an ideal gas.

(a) Find the pressure of the gas after being compressed. (2 marks)

(b) (i) Find the average kinetic energy of the gas molecules. (2 marks)

(ii) The mass of one mole of neon gas is 0.020 kg . Find the root-mean-square speed of the gas molecules. (2 marks)

Answers written in the margins will not be marked.

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Book 1 section 2 Heat Capacity: DSE long questions

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